Quiz # 4

Course Title: Software Construction and Development

Name: _____ Registration No.: _____

Section: _____ Date: _____ Total Marks: _____

CLO#	Course Learning Outcome (CLO)	Taxonomy Level
CLO 2	Apply software design principles, refactoring techniques and design patterns to develop collaborative, maintainable, and high-quality Java applications.	C3

Question 1:

Analyze the following software development scenario and propose a complete solution.

Scenario: A development team is working on a Java-based online booking system. Developers manually download libraries, different team members use different dependency versions, and every release is built and deployed manually. The application processes 6,000 booking records in 30 seconds. Repeated calculations are performed inside loops, unnecessary objects are continuously created, memory usage keeps increasing, and multiple threads sometimes update the same booking count incorrectly.

Required Tasks:

- Recommend Maven or Gradle for this project. Justify your choice and explain how it will manage dependencies and solve version conflicts.
- Design a suitable CI/CD pipeline for the team. Write its stages in the correct order and explain what happens at each stage.
- Calculate the current performance in records per second. After optimization, the system processes the same 6,000 records in 12 seconds; calculate the new performance and compare both results.
- Identify and explain four suitable code-optimization techniques that should be applied to the given application.
- Explain how performance profiling can help locate CPU and memory bottlenecks, and suggest a solution for the continuously increasing memory usage.
- Identify the concurrency problem affecting the booking count and explain how synchronization can solve it.

Support each recommendation with a clear reason and use relevant examples where necessary.